

OPERATOR'S MANUAL

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CHAPTER TWO DESCRIPTION AND SPECIFICATIONS

Description

The Infinity™ Nd:YAG laser system consists of an internal diode pump, Q-switched, master oscillator which provides the seed pulse for a dual rod, single lamp, amplifier stage. An overview of the Infinity's three primary assemblies is given in Figure 2-1. Also illustrated in Figure 2-1 is the suggested setup orientation for each of the components.

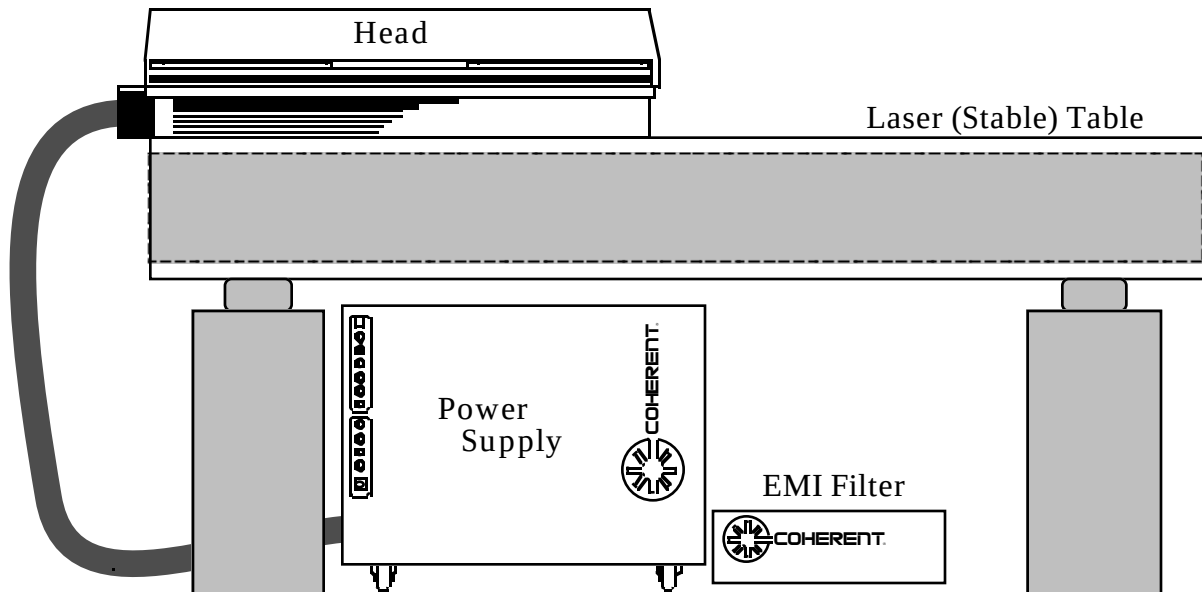


Figure 2-1. Infinity Main Assemblies.

Note that the external EMI filter (under table, lower right of figure) is only part of systems which are shipped to the European Community. This component is wired between the facility power the Infinity power supply.

An optical schematic of the Infinity laser head is presented in Figure 2-2. Refer to Chapters Three and Eight for complete descriptions of each component in the Infinity Laser System.

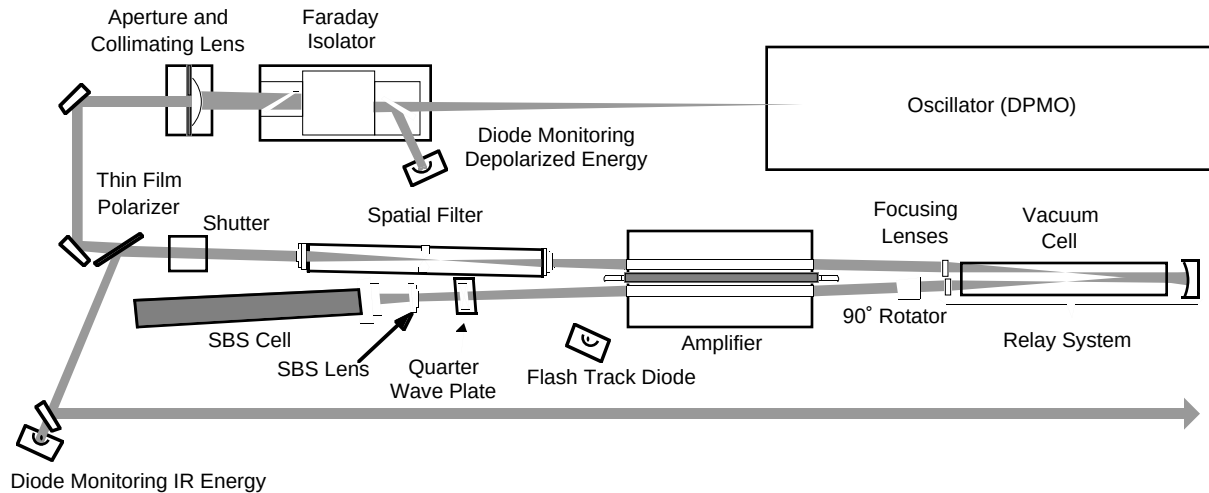


Figure 2-2. Infinity Optical Schematic.

Pulse Energy Servo

The Infinity is equipped with a feedback system that improves long term stability of the laser output. This system utilizes the output from a photo-diode measuring the IR energy to adjust the voltage across the capacitor bank. This feature is enabled or disabled in the computer interface. When the output power is set by choosing the voltage applied to the flash lamp, the light regulation system is disabled and when it is set by choosing mJ/pulse it is enabled. Operating the system using the lamp voltage control is normally a service procedure. See Chapter Seven, System Software, for complete details.

Safety Interlocks

In order to protect the user(s), four mechanical safety interlocks and a mechanical shutter have been built into the system. The interlocks can be tripped by opening the laser head cover, opening the lid covering the flash lamp leads, or removing one of dowel pins adjacent to the amplifier assembly latches. In addition there is an external interlock connector at the rear of the power supply. This provides the user with the option of installing a safety interlock switch connected to an external device such as a lab door.

If any of the interlocks are tripped while the laser is running, all systems except the water pump will be switched off immediately and the fault message: "Interlock open" displayed (obviously interlocks inside the laser head can only be tripped if the head cover interlock first has been defeated). Before the laser can be restarted, the off button needs to be pressed and the closed position of the interlocks restored. Thereafter the normal start-up procedure, as described in Chapter Four, can be used. If the fault message is cleared but

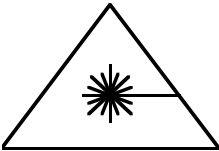
the fault condition still exists, the message will reappear within a few seconds.

If the laser is operated with the laser head cover interlock defeated, and the cover is either opened or closed while the laser is running, the beam will strike the inside of the cover while it is being translated through the beam path. This may give rise to sputtering which can deposit onto the optics resulting in irreversible damage.

The electromagnetically activated shutter is positioned in the beam path between the thin-film polarizer and the spatial filter. It provides a way of preventing a laser pulse from leaving the laser head without changing the thermal load on the Nd:YAG rods.



Never open or close the laser head cover while operating the laser with the interlock defeated. Irreversible damage may occur to optical components.



Operating the laser with the interlocks defeated is a service procedure. It should only be carried out by a qualified technician who is well aware of the potential hazards. Damage that occurs to the system while operated with defeated interlocks is not covered by Coherent warranty.

PulseGuard™

PulseGuard is a safety system designed to protect the laser by continuously monitoring the depolarized light intensity. Depolarized light is laser energy that after amplification, instead of being reflected out of the laser by the thin film polarizer, is transmitted through it and directed back into the Faraday Isolator. It is desirable to keep this energy component less than or equal to 1.5% of the output energy at the systems maximum repetition rate and pulse energy. If the energy going through the Faraday Isolator, as measured by the photodiode attached to it, exceeds 8 mJ/pulse the on board microprocessor will shut off the system. This is done in order to protect the master oscillator from being damaged. A bar graph display showing the depolarized energy is available in the Infinity system “Monitor” window. This window can be accessed by clicking on the “Monitors...” button found in the main Infinity software window. See Chapter Seven, System Software, for additional information.

FlashTrack Servo Controls

FlashTrack is a control system that reduces shot-to-shot intensity fluctuations. A photo-diode monitors the emission intensity of the flash lamp during the first half of the lamp pulse. A feed-forward circuit then uses this information to make a small correction to the lamp current during the second half of the lamp pulse such that the total amplification is constant over time. The principle of operation is illustrated in Figure 2-3. A photo-diode connected to an integrator monitors the flash lamp emission. After half of the lamp width time has elapsed a sample and hold circuit reads out the integrator and converts it to a voltage. This voltage is summed with V_{Coarse} and V_{Vernier} (both inverted) and then amplified. The amplified signal is converted to a pulse that has fixed height and a length proportional to the voltage. This pulse is used to drive a transistor switch regulating the current going through the lamp. If the first half of a lamp pulse is brighter than average, the amplifier output goes down (signal applied to negative input of amplifier). This yields a shorter drive pulse for the transistor switch which decreases the amount of current going through the lamp during the second half of the lamp pulse. See Chapter Seven, System Software, for additional information.

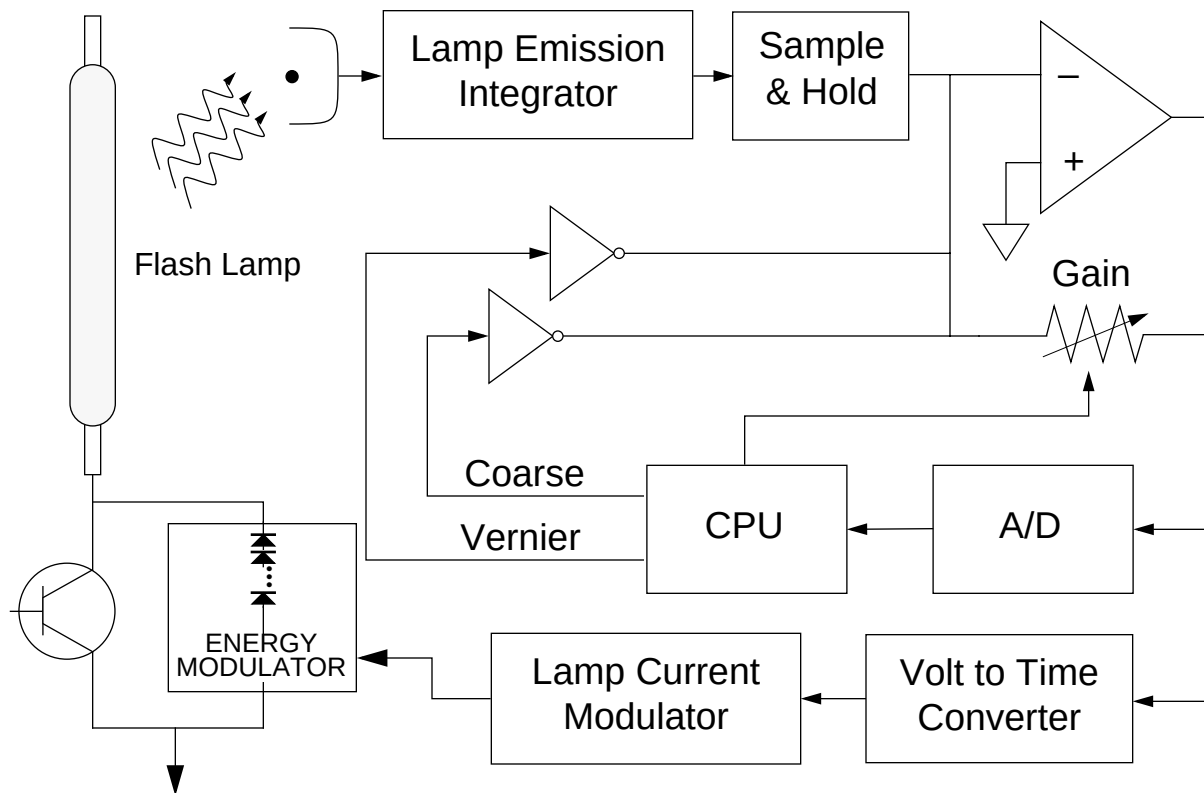


Figure 2-3. Block Diagram for FlashTrack Servo.

Water Cooling System

For every laser pulse of a 100 Hz system, about 40 Joules of energy is deposited into the flash lamp. This means that 4000 W are deposited into the amplifier head. Only a small fraction of this energy is converted into laser light. Because of the large heat load, the small dimensions, and the stringent requirements, an internal water cooling loop is used. This water is cooled using externally supplied water in a heat exchanger located inside the power supply. The layout of the water cooling system is illustrated in Figure 2-4.

Tap water enters the power supply and goes through a heat exchanger and then back out to the drain. The internal water goes (after being cooled in the heat exchanger) up to the laser head, where it cools off the flash lamp, the laser rods, and some of the electronics located under the base plate. After this the water flows back to the power supply where it goes through a water filter with 10 μm pore size, followed by an instrument that monitors the water resistivity. If the conductivity of the water rises above or below preset values, the computer will display a fault message. In order to keep the water deionized, a fraction of it is sent through a deionization filter while the main flow is directed back to the main reservoir. The internal cooling water is also used to regulate the temperature of some of the electronics in the power supply, see Figure 2-4.

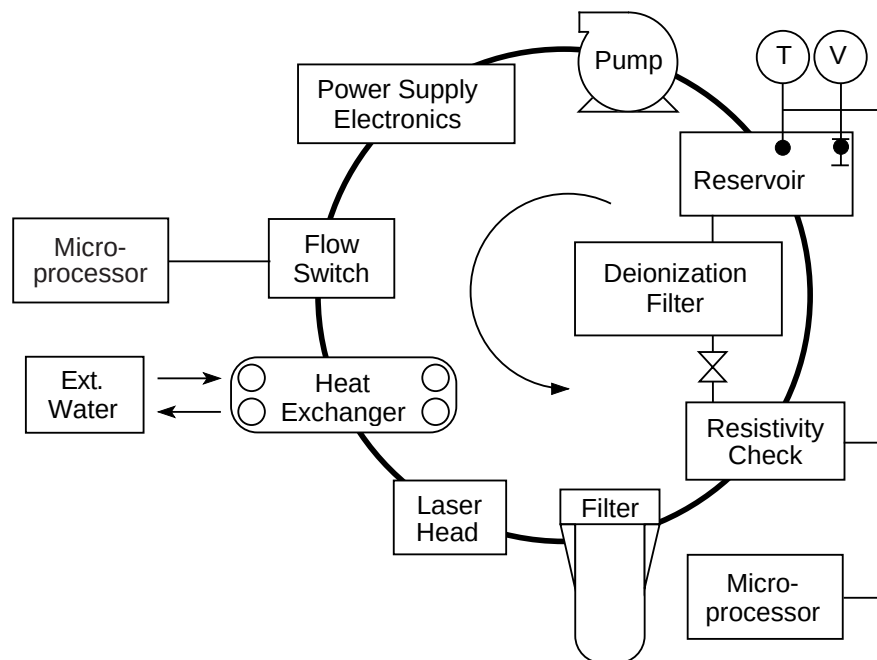
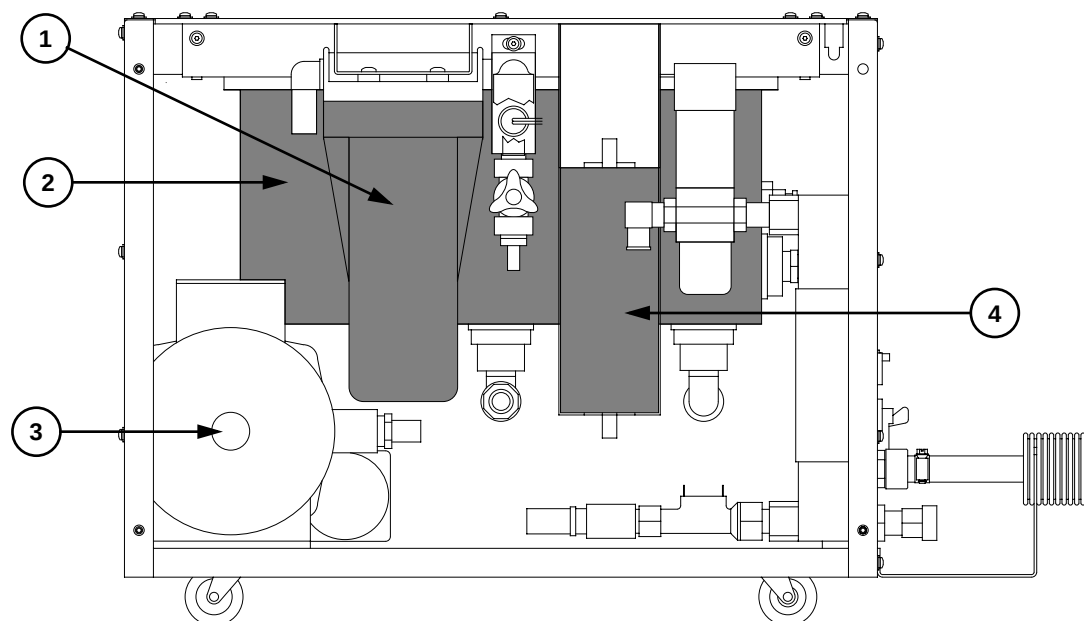


Figure 2-4. Schematic of Infinity Water Cooling Loop.

Heat Exchanger

It is recommended that the internal cooling water, the deionization filter, and the particle filter be changed once every year. Major heat exchanger components, the right hand side of the Infinity power supply, are illustrated in Figure 2-5.



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|--------------------|---------------------------|
| 1. Particle Filter | 3. Three Phase Water Pump |
| 2. Water Reservoir | 4. Deionization Cartridge |

Figure 2-5. Major Heat Exchanger Components.

Infinity Specifications

Table 2-1. Infinity Pulse Energy Specifications

	PULSE ENERGY AT [mJ/pulse]			
REP. RATE	1064 nm	532 nm	355 nm	266 nm
0.1 - 20 Hz	600	300	240	50 ¹
20 - 50 Hz	500	250	200	40
50 - 100 Hz	400	200	160	25

¹ Pulse energy at 266 nm is 50mJ/ pulse at 10 Hz.

Table 2-2. Infinity Noise Specifications

	NOISE AT [% std. dev]			
REP. RATE	1064 nm	532 nm	355 nm	266 nm
0.1 - 20 Hz	1.0	1.8	2.0	<2.5
20 - 50 Hz	1.0	1.8	2.0	<2.5
50 - 100 Hz	1.7	2.5	2.5	<4.0

Table 2-3. Infinity Typical Pulse Widths

WAVELENGTH [nm]	PULSE WIDTH [ns]
1064	3.5
532	3.0
355	3.0
266	2.5

Table 2-4. Infinity System Specifications

PARAMETER	SPECIFICATIONS
IR mode quality	< 1.5x Diffraction limit
Beam diameter, near field ¹	> 95% of pulse energy through 5.5 mm at SHG crystal
Beam diameter, far field	> 84% of pulse energy into < 0.7 mrad divergence
Divergence, full angle	< 0.7 mrad
Linewidth Single-longitudinal mode Single shot (typical)	< 2x transform limit < 250 MHz
Timing jitter (rms) rel. Q-switch synch	< 500 psec

¹ Near-field image is formed at the SHG crystal position.